Comparative Analysis on Self-Supervised Automatic Speech Recognition Techniques for Low Signal-to-Noise Ratios

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***Abstract*—Automatic Speech Recognition (ASR) uses artificial intelligence to convert speech into text. Self-supervised learning techniques using transformers can perform ASR on unlabeled datasets, such as the Wav2Vec2 model. The performance of the model is heavily reliant on the training data, and can perform poorly in noisy environments. This paper explores a comparative analysis on performance degradation of various signal to noise ratios (SNR) on the model using improvements such as adversarial training, and Large Language Model (LLM) Post-Processing individually and in combination. Wav2Vec2 will be fine-tuned and evaluated with multiple SNR ratios, using these improvements to determine which combination yields the lowest Word Error Rate (WER).**

***Index Terms*—Machine Learning, Automatic Speech Recognition (ASR), Large Language Model (LLM), Signal to Noise Ratio (SNR), Word Error Rate (WER), Adversarial training**

I. Introduction

Automatic Speech Recognition is an AI solution aiming to convert speech into text. It is most prominently used in social media and virtual meetings to transcribe what is being said into captions for a viewer to read, but has many real world applications. Self-Supervised ASR such as Wav2Vec2 improves upon past models using encoders and transformers, which allows a model to learn from unlabeled data by learning certain representations from the structure of the speech before being fine tuned with a smaller labeled dataset [2]. This important advancement reduces the need to generate large labeled datasets, an often timely and costly procedure, only requiring small amounts of labeled data for fine-tuning. However, self-supervised ASR experiences performance degradation from noisy or unclean audio [1]. The purpose of this research is to determine a pipeline to improve performance of speech with low SNR. This study hypothesizes that combining Noise augmentation, Adversarial Training, and Large Language Model Post-Processing will reduce the Word Error Rate of low SNR speech over any single technique alone. Analysis of each method along with their performance on various SNR thresholds will also help categorize where each method struggles, and provide information for further research as to what still needs improvement.

II. Related Work

Baevski and Alexei's "WAV2VEC 2.0: A Framework for Self-Supervised Learning of Speech Representations" [2] explores the performance of self-supervised transformer based ASR compared to supervised learning techniques. The model has a lower WER than standard supervised and semi-supervised learning techniques, demonstrating the impact of self-supervised learning. These results are on clean librispeech data only and doesn't explore noisy conditions.

Wav2code: Restore Clean Speech Representations via Codebook Lookup for Noise-Robust ASR [7] explores performance degradation on Wav2Vec2 and other models at various SNR ratios. It shows how the model fine-tuned on cleaned data degrades as noise decreases, with a 66.5% WER at 0dB and 28.2% WER at 20dB in metro settings. This shows the motivation for this paper, exploring the changes of degradation with different methods of noise mitigation.

Masked Language Model Scoring [8] demonstrates the abilities of improving ASR performance through rescoring. The paper paper shows improvement of Librispeech model's WER by 30% relative. Using this paper's ideas, this research will explore using a large language model as a post processing step to fix noise related errors.

In Comparative Study on Noise-Augmented Training and Its Effect on Adversarial Robustness in ASR Systems [9], Pizzi discovers that noise-augmented training has little effect on Wav2Vec2 performance. This shows Wav2Vec2's resistance to adversarial attacks, which yields that noise augmented fine-tuning may have no impact on WER. However PGD-based adversarial training may still improve performance over normal augmentation. These findings also show that it is important to explore re-scoring in conjunction with adversarial training for noisy data as adversarial training alone may not have strong impact on WER.

III. Research Project Problem

Self-Supervised ASR experiences performance degradation with audio containing background noise and other fluctuations. Similar research has proposed solutions such as adversarial training, and LLM post-processing, but these papers isolate and explore a single solution mainly on clean test sets [3]. The overall impact of these methods on low SNR data is relatively unexplored. This research provides a comparison of each method individually, along with a combination of all methods on a standardized test dataset, examined across multiple signal-to-noise ratio thresholds. Such a comparison is important in furthering research by determining how each method performs across the various thresholds.

IV. Method

A. Dataset

This paper uses Librispeech dev-clean dataset for fine-tuning with augmented data and adversarial training, as Wav2Vec2 is already pretrained and fine-tuned on 960 hours of data. It is 5.4 hours of 16kHz data containing 2703 utterances. It contains english speech derived from read audiobooks with transcriptions of each sentence [4]. Librispeech test-clean is used for the test dataset, containing 2620 utterances from the same sources. To augment the data with noise at various SNR levels, the WSJ0 Hipster Ambient Mixture (WHAM!) dataset is used which contains environmental noise such as restaurants, cafes, bars, and parks [5].

B. Feature Extraction

Wav2Vec2 operates on raw audio, rather than engineered features. The audio is converted to 16kHz and normalized before training and inference.

C. Model

The ASR model used for this paper is the Wav2Vec2 pretrained using self-supervised learning and fine-tuned on 960 hours of the Librispeech train Dataset [6]. Wav2Vec2 uses Connectionist Temporal Classification (CTC) loss to learn transcriptions without providing timestamps for each word. The pretrained model is used as a baseline and will be used for fine-tuning. Figure 1 and 2 show the baseline and fully implemented pipeline.

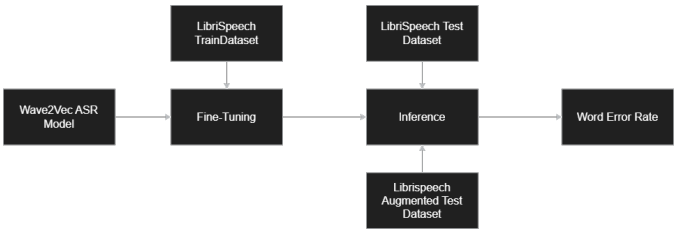


Fig. 1. Baseline ASR Architecture

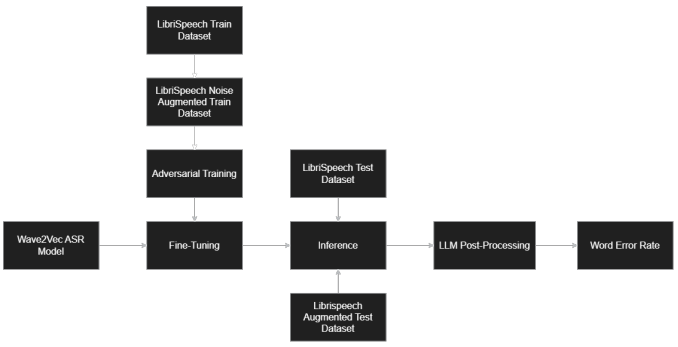


Fig. 2. Full ASR Pipeline with Adversarial Training and LLM Post-processing

D. Noise-Augmented Fine-Tuning

The model was fine-tuned with LibriSpeech's 5 hours of dev-clean utterances augmented with the WHAM! dataset to adapt to noisy conditions. The utterances are augmented randomly at -5dB, 0dB, 5dB, and 10dB SNR.

E. Adversarial Training

Adversarial training is performed to generate utterances that increase the model's loss, improving it's learning and robustness. The generated examples are created using Fast Gradient Sign Method (FGSM), which manipulates the audio in the direction of the gradient that increases CTC loss. Equation 1 shows how each input is manipulated.

$$x_{\text{adv}} = x + \epsilon \,\text{sign}\left(\nabla_x \mathcal{L}(\theta, x, y)\right) \tag{1}$$

F. LLM Rescoring

GPT2 is used as the large language model to rescore the ASR model's predictions. GPT2 creates a language model score for multiple candidate transcriptions from a beam search. The language model score is combined with the CTC score to generate the final prediction. Formula 2 shows the calculation, with alpha scaling the impact of the language model score.

$$S_{\text{combined}} = S_{\text{CTC}} + \alpha S_{\text{LM}}, \tag{2}$$

G. Evaluation

The metric used for evaluation is Word Error Rate (WER). This is calculated from a comparison of the model's output to the ground truth. Formula (3) shows how WER is calculated, where S is the substitutions or incorrect words, D is deletions or missing words, I is the insertions or extra words, and N is the total number of words in the ground truth.

$$WER = \frac{S + D + I}{N} \tag{3}$$

The WER will be calculated on the clean test dataset as well as at the dataset augmented at SNR ratios -5dB, 0dB, 5dB, 10dB and is calculated from formula (4).

$$SNR_{dB} = 10log_{10}\frac{P_{signal}}{P_{noise}} \tag{4}$$

V. Experiment

The experiment consists of comparing the baseline model with three fine-tuned models with and without LLM re-scoring on clean data and various SNRs. The three models were separately fine-tuned using the dev-clean data with noise augmentation, FGSM adversarial examples, and a combination of noise augmentation and adversarial examples. The WHAM! dataset was used for noise augmentation selecting one noise from each location and augmenting at SNRs from -5, 0, 5, and 10 dB. FGSM was applied with $\epsilon = 0.001$. In the combination model, adversarial examples were created from the noise augmented data before fine tuning.

TABLE I
WORD ERROR RATE (WER) ACROSS CLEAN AND NOISY TEST CONDITIONS. LOWER WER INDICATES BETTER RECOGNITION PERFORMANCE.

SNR	Baseline	Baseline + LM	Noise FT	Noise FT + LM	Adv. FT	Adv. FT + LM	Noise + Adv. FT	Noise + Adv. FT + LM
Clean	17.02%	16.75%	5.19%	4.92%	4.65%	**4.26%**	5.67%	5.43%
-5 dB	84.99%	87.71%	**68.41%**	69.45%	77.04%	78.24%	69.58%	70.07%
0 dB	66.84%	69.29%	**41.09%**	42.81%	51.05%	50.59%	42.16%	43.54%
5 dB	43.90%	47.38%	**19.54%**	20.67%	23.43%	24.13%	19.93%	20.74%
10 dB	28.96%	30.61%	**10.15%**	10.15%	10.97%	10.86%	10.59%	10.63%

Each model was trained for 1 epoch with a batch size of 4 using AdamW optimizer with a learning rate of 1×10^{-5}. The feature encoder is frozen for each model.

Each model was evaluated with LibriSpeech test-clean with both clean and noise-augmented data using the WHAM! dataset at -5, 0, 5, and 10dB. This evaluation data was used to calculate the WER of each model.

GPT-2 was used as the LLM to perform rescoring, using a CTC beam search to generate the top 5 candidate transcriptions. $\alpha = .1$ was chosen for generating the combined score and final prediction.

VI. RESULTS

Table 1 shows the WER from each configuration. Fine-tuning shows reduced error rate compared to the baseline model, with the best performance from the noise-augmented fine-tuned model. This performance reached WERs of 68.41%, 41.09%, 19.54% and 10.15% on SNRs of -5, 0, 5, and 10dB. Compared to the baseline of 84.99%, 66.84%, 43.90%, and 28.96% , this shows substantial improvements in noisy conditions. Adversarial fine-tuning and the combination of noise-augmented and adversarial fine-tuning performed better than the baseline but did not outperform the purely noise-augmented model.

Large Language Model re-scoring had little to no effect on improving performance in noisy conditions. It did however achieve the strongest performance on the clean dataset with the adversarial fine-tuned model at 4.26% WER compared to the baseline of 17.02%.

Overall, the noise-augmented model performed the best when dealing with noisy conditions, and adversarial training had the best performance on clean data. Using the combination model with LLM rescoring did not reach better performance than the purely noise-augmented model.

VII. FUTURE IMPROVEMENTS

In order to fully explore the impact of these changes on WER in noisy conditions, some further exploration must be done. Further hyperparameter tuning with the FGSM ϵ and LLM rescoring α should be done to determine if there are better values for each. Denoising on the noise-augmented test dataset should be explored to determine if performance improves. Additional large language models should be experimented with as more recent models may improve the performance. Finally, time constraints limited the amount of fine-tuning which was only done for one epoch. The models may not have fully converged, limiting the validity of the conclusions drawn. Further experiments with more epochs should be pursued.

VIII. CONCLUSION

This research found that the combination of noise-augmented and adversarial fine-tuning with large language model re-scoring does not improve performance in noisy conditions over fine-tuning with a noise-augmented dataset alone. Exploring the impact of these improvements alone and in combination is important to determine whether they can improve ASR robustness. The purely noise-augmented fine-tuned model performed the best in noisy conditions, but other combinations such as different language models or denoising may increase performance further. Although adversarial fine-tuning and the combination did not reach higher performance than the noise-augmented dataset, these models had high performance on the clean dataset and still performed better than the baseline model. Overall, noise-augmented fine-tuning is currently the most effective technique for ASR in noisy conditions among the techniques explored in this paper.

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